

Regularized and Stable Lexical Classification in High-Dimensional Semantic Feature Spaces

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Based on Testa (2026) · Binder et al. (2016)

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Research Question & Dataset

Can we predict the grammatical word class from primitive semantic features?
Does high-dimensional expansion further improve classification?

535

Lexical entries
(nouns, verbs, adjectives)

65

Semantic feature
dimensions

434 / 62 / 39

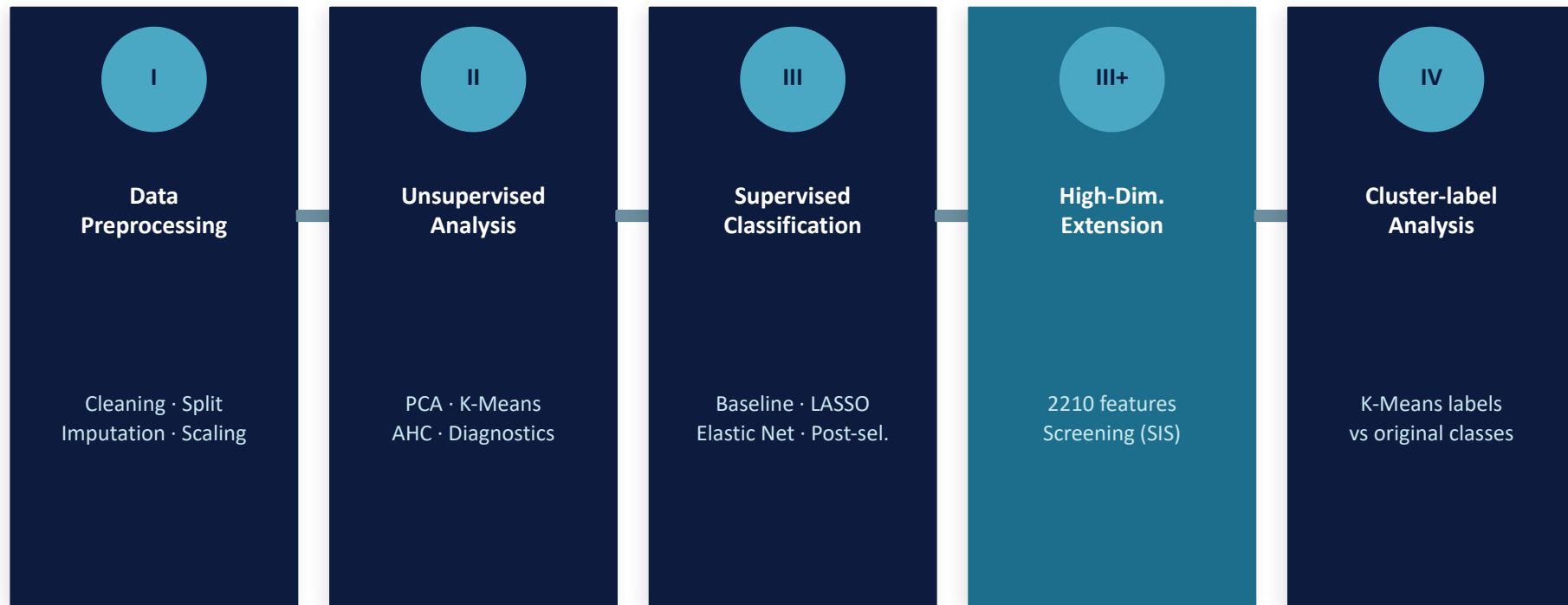
Thing / Action / Property
(81% / 12% / 7%)

428 / 107

Train / Test
(stratified 80/20)

△ *Class imbalance: a dummy majority classifier (always predict noun) achieves 81.3% accuracy — macro-F1 only 0.299. Balanced accuracy and macro-F1 are therefore the primary evaluation metrics.*

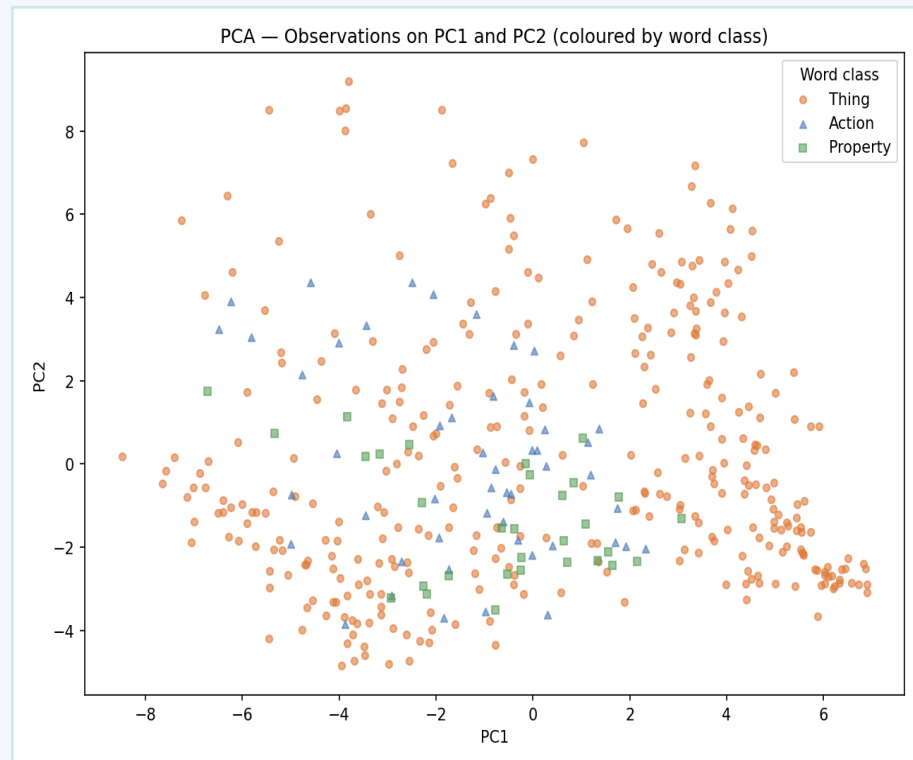
Analysis Pipeline



Extra: Alternative classifiers (SVM · Random Forest · k-NN) — same selected feature space

PCA: Variance Structure

- ▶ PC1 explains 22.7%, PC2 explains 13.4% — combined 36.1%
- ▶ 10 PCs needed to reach ~75% cumulative variance
- ▶ PC1: sensorimotor gradient (VISION, COLOUR, SHAPE, TOUCH, TEXTURE) — concreteness axis
- ▶ PC2: auditory/affective (FAST, SOUND, PAIN, ATTENTION, LOUD) — dynamic/eventive axis
- ▶ Broad but imperfect class separation — boundaries not sharp



PCA Biplot — PC1 vs PC2 (coloured by word class)

Clustering: K-Means and AHC

Key finding: clusters do NOT align with grammatical classes · K-Means K=3: ARI=0.030, NMI=0.072 · AHC K=3: ARI=-0.067, NMI=0.042

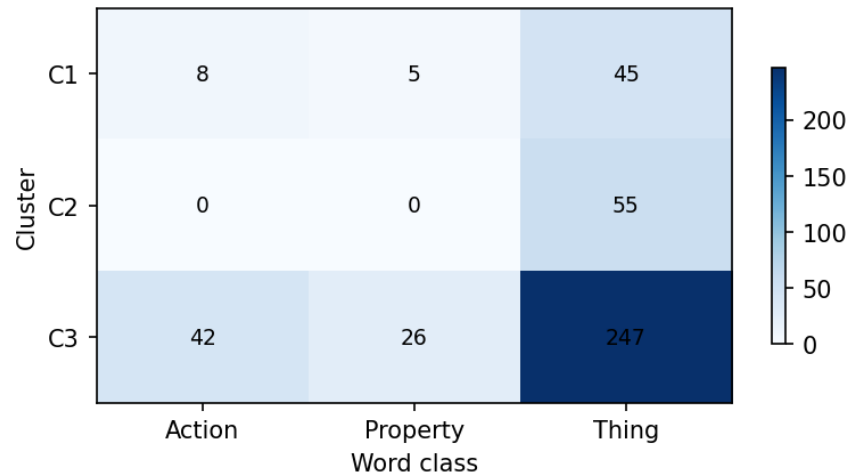
K-Means K=3 vs original labels

Cluster vs label — kmeans k=3
ARI=0.030 NMI=0.072



AHC K=3 vs original labels

Cluster vs label — ahc k=3
ARI=-0.067 NMI=0.042



Supervised Baseline

Unpenalised multinomial logistic regression on all 65 standardised features

95.3%

Test Accuracy

Train: 100% (perfect sep.)

91.9%

Balanced Accuracy

All 3 classes recovered

0.890

Test Macro F1

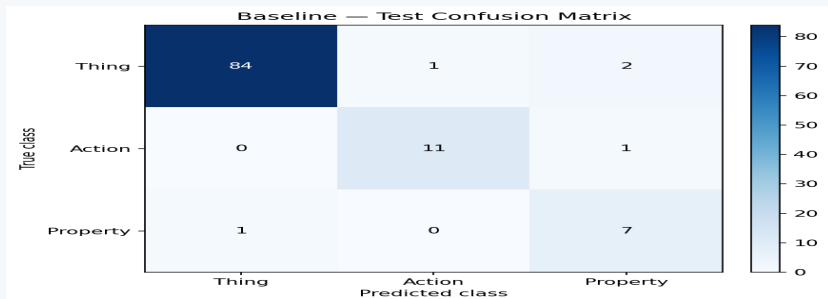
CV mean: 0.859 ± 0.045

81.3%

Dummy Majority

Macro F1 = 0.299 only

Test Confusion Matrix



Per-class breakdown

Thing (noun)	Excellent	84/87 correct — dominant class
Action (verb)	Good	11/12 correct — motion/social features key
Property (adj.)	Good	7/8 correct — fewest obs. (7%)

Multicollinearity in the Semantic Space

199

Pairs $|r| \geq 0.50$

22

Pairs $|r| \geq 0.80$

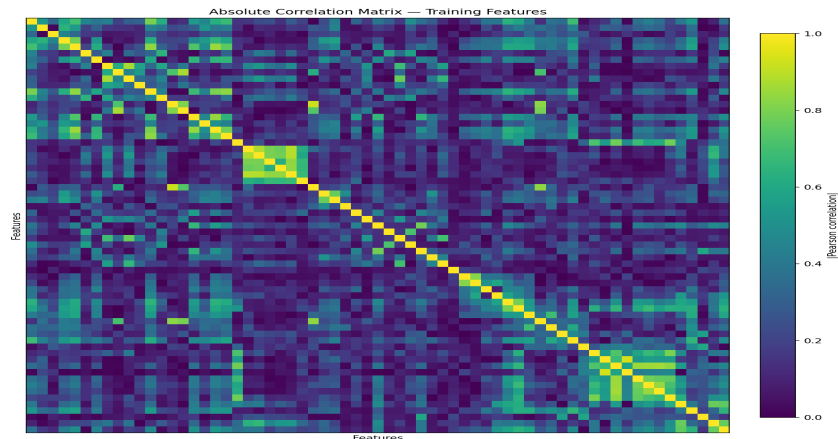
0.937

Max $|r|$
(pleasant–happy)

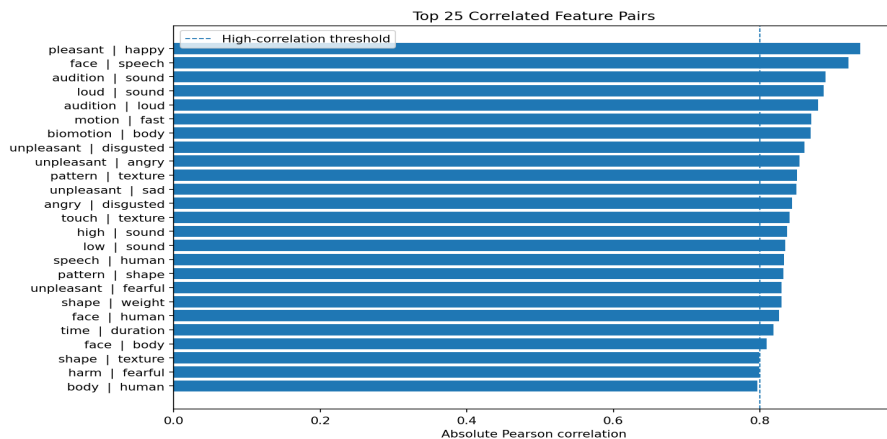
51

Features VIF ≥ 5
(max VIF: 23)

Absolute Correlation Heatmap



Top Correlated Feature Pairs



Redundancy is structural, not accidental — it reflects the co-activation of semantic domains in word meanings. Motivates LASSO and Elastic Net.

Regularised Models: LASSO & Elastic Net

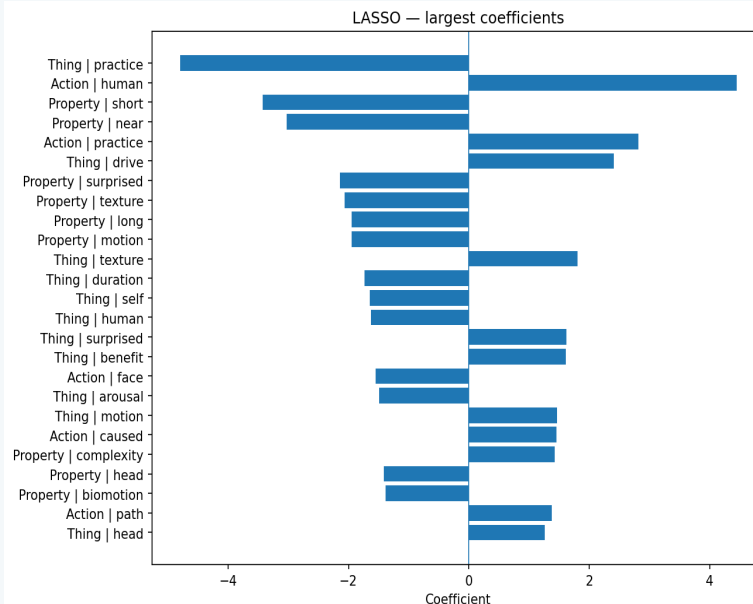
LASSO (L1)

- Best $C = 100$ (weak regularisation needed)
- 172 / 195 non-zero coefs (zero share 11.8%)
- All 65 features active — no full sparsity at $p=65$
- Test Macro F1 = 0.890 ·
Balanced Acc. = 0.919

Elastic Net (L1+L2)

- Best $C = 100$, L1 ratio = 0.3
- 192 / 195 non-zero coefs (zero share 1.5%)
- L2 dominates — nearly all features retained
- Test Macro F1 = 0.890 · same as LASSO

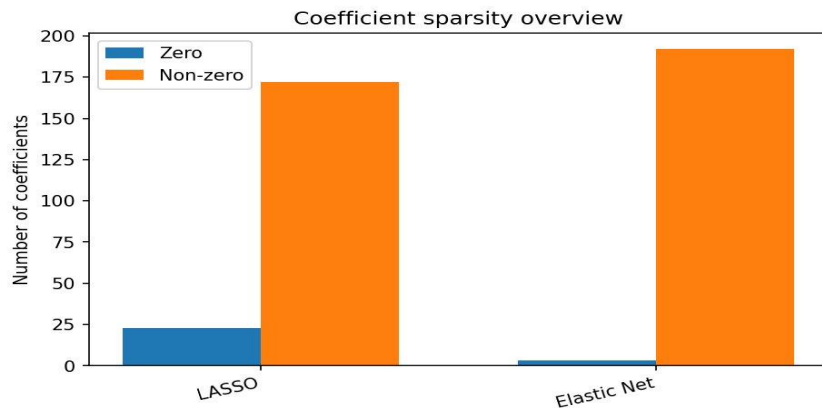
Top LASSO coefficients



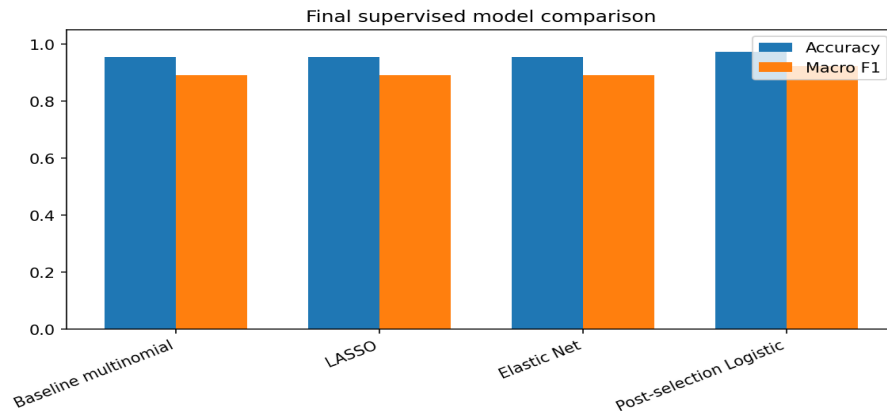
Note: paper body Table 7 reports higher values (LASSO 0.968/0.915, EN 0.971/0.920) — paper appendix Table 24 and actual code output both confirm 0.953/0.890.

Post-Selection Model & Final Comparison

Sparsity overview



Model comparison (test set)

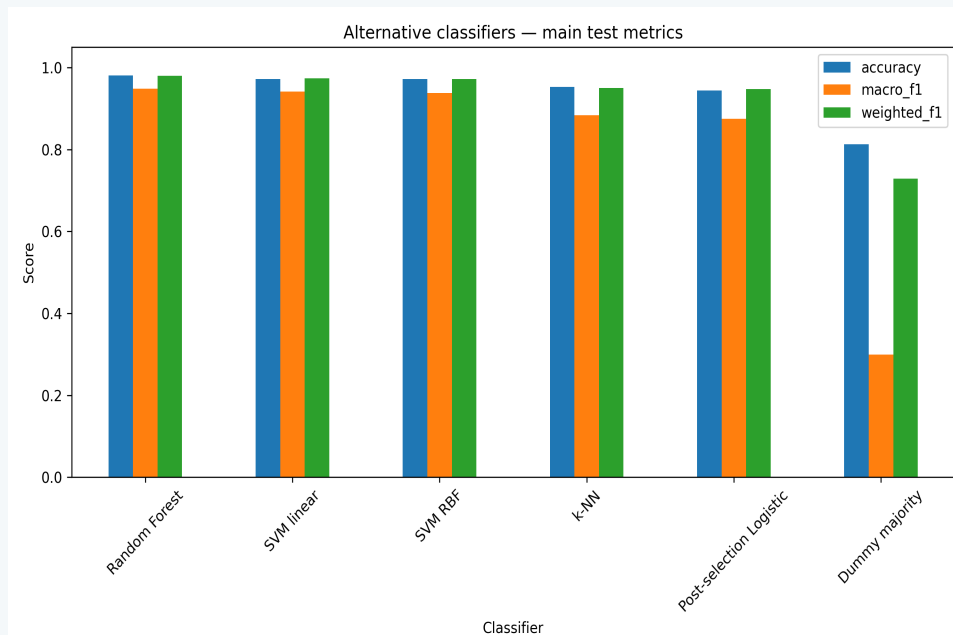


Model	Acc.	Bal.Acc.	Macro F1
Baseline multinomial	0.953	0.919	0.890
LASSO	0.953	0.919	0.890
Elastic Net	0.953	0.919	0.890
Post-selection Logistic	0.972	0.927	0.923

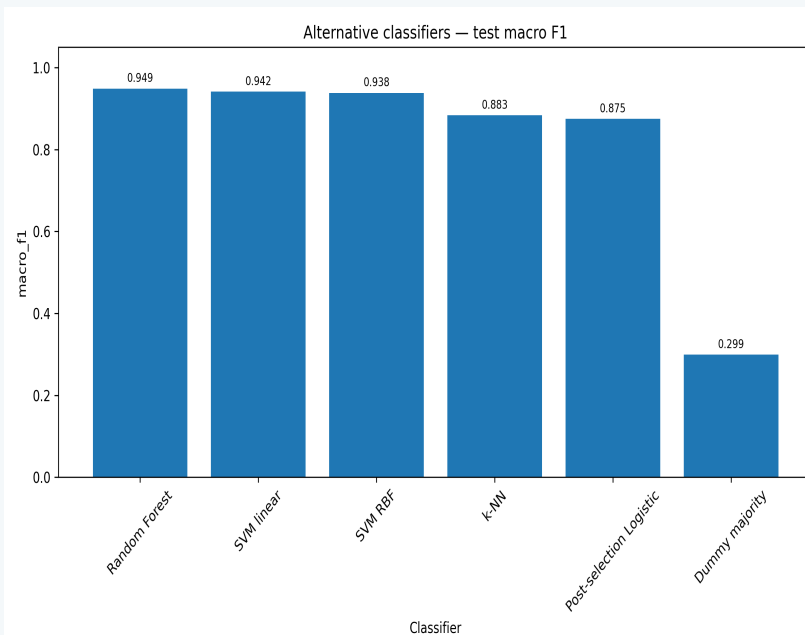
Alternative Classifiers Comparison

Same 65 selected features — does the classifier family matter?

Main test metrics



Macro F1 ranking



RF: F1=0.949 (best accuracy) · SVM linear: bal.acc.=0.989 (best minority recall) · SVM RBF: C=1, $\gamma=0.01$ · k-NN: k=3, euclidean, uniform

High-Dimensional Expansion: 2210 Features

- ▶ 65 original + 65 squared + 2080 pairwise interactions = 2210 features
- ▶ $p/n = 2210/428 \approx 5.16 \rightarrow$ genuine $p \gg n$ regime
- ▶ $X'X$ is singular: rank deficiency ≥ 1782 — classical VIF undefined
- ▶ 127,467 pairs with $|r| \geq 0.50$; 1,945 pairs with $|r| \geq 0.90$

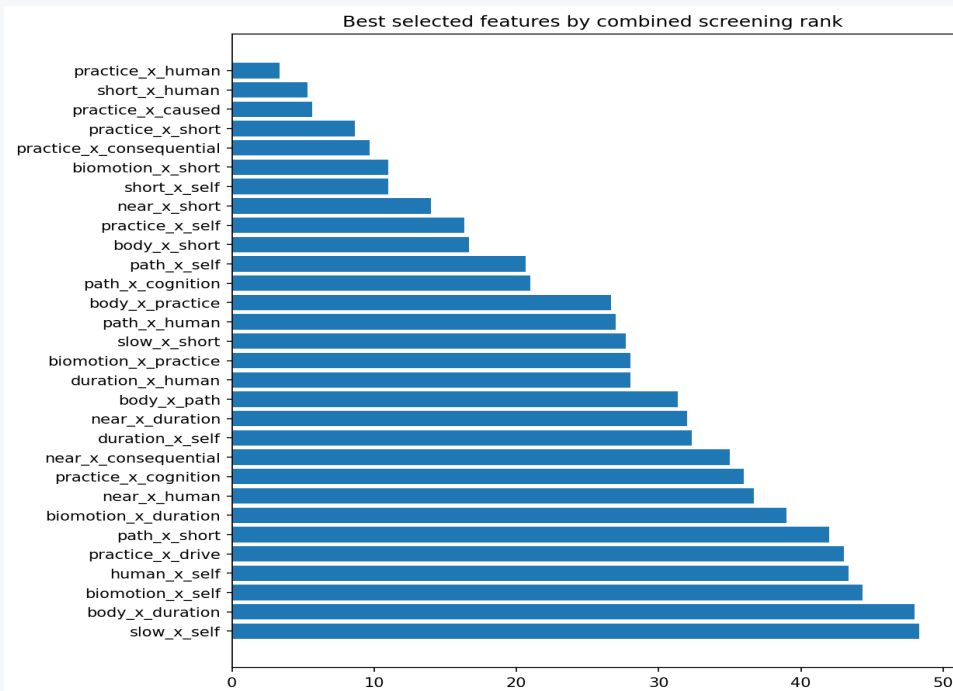


65 vs 2210 features		
	65	2210
$p > n$	No ($p < n$)	Yes ($p \gg n$)
VIF	Computable max=23	Undefined ($X'X$ singular)
Screening	Not needed	SIS $\rightarrow$ top 300
Ridge baseline	Acc. 0.953 F1 0.890	Acc. 0.972 F1 0.929
LASSO/EN selected	65 (all)	234 / 253
Best test F1	0.923	1.000*

* Perfect test F1=1.000 on HD — small test set (12 verbs, 8 adjectives), interpret cautiously

Feature Screening on 2210 Features

Combined screening (ANOVA F-score + Mutual Information + class-mean contrast) → top 300 retained



Top 10 screened features

- #1 practice_x_human
- #2 short_x_human
- #3 practice_x_caused
- #4 practice_x_short
- #5 practice_x_consequential
- #6 biomotion_x_short
- #7 short_x_self
- #8 near_x_short
- #9 practice_x_self
- #10 body_x_short

Supervised Analysis with K-Means Labels

What if we use the K-Means K=3 clusters as target labels instead of the original word classes?

Cluster	Train	Test
Cluster_01	67 (15.7%)	21 (19.6%)
Cluster_02	183 (42.8%)	34 (31.8%)
Cluster_03	178 (41.6%)	52 (48.6%)

Cluster-label model performance (test set)

Model	Acc.	Bal.Acc.	F1
Baseline	0.981	0.974	0.979
LASSO	0.972	0.958	0.967
Elastic Net	0.981	0.974	0.979
Post-sel. (53 feat.)	0.981	0.974	0.979

Interpretation

- ▶ Clusters are more balanced than original labels — easier to classify
- ▶ High F1 (0.979) confirms the semantic feature space encodes cluster structure well
- ▶ Original labels (Thing/Action/Property): best F1 = 0.923
- ▶ Cluster labels: F1 = 0.979 — semantic space fits its own clusters better than linguistic categories
- ▶ Confirms: grammar $\neq$ semantic similarity, but both are grounded in the Binder features

Key Findings

1

Semantic features are highly predictive

Post-selection logistic: 97.2% accuracy, Macro F1 0.923.
Far above the 81.3% dummy majority (F1=0.299).

3

Random Forest leads; linear models competitive

RF: F1=0.949. SVM linear: best balanced acc. (0.989).
Post-sel. logistic: F1=0.923, fully interpretable.

2

Clusters $\neq$ grammatical classes

K-Means K=3 ARI=0.03, AHC ARI=-0.07. Semantic space captures fine-grained domains, not grammar.

4

Interaction terms are the strongest HD predictors

practice_x_human, short_x_human, biomotion_x_short
dominate screening. Class info lies in semantic combinations.

Conclusions

- Primitive semantic features reliably predict grammatical word class: accuracy 95.3% (baseline) → 97.2% (post-selection), Macro F1 0.890 → 0.923.
- Clustering does NOT recover grammatical structure ($ARI \approx 0.03$ / -0.07) — the semantic space encodes conceptual similarity, not morphosyntax.
- Regularisation does not induce sparsity at $p=65$: LASSO/Elastic Net keep all features active. Post-selection L2 refit provides the best final linear model.
- Random Forest ($F1=0.949$) is the best single classifier; SVM linear achieves the best balanced accuracy (0.989). Linear models remain strongly competitive.
- HD expansion (2210 features, $p \gg n$) with screening + regularisation exposes interaction-based semantic structure. Strongest predictors are semantic combinations (practice_x_human, biomotion_x_short...).

Thank you